Tutorial 3

1. A number is selected uniformly at random from the interval

$$\left(0, \frac{\pi}{2}\right).$$

   What is the probability that its sine is greater than its cosine?

2. Can a function of the form

$$p_X(x) = \begin{cases} c\left(\frac{2}{3}\right)^x, & x = 1, 2, 3, \ldots, \\ 0, & \text{elsewhere,} \end{cases}$$

   be a probability mass function? If so, determine the value of c.

3. The number of wrong-number phone calls received by a certain mail-order house follows a Poisson distribution. On average, the mail-order house receives 7 wrong-number calls per week.

   What is the probability that they will receive

   (a) exactly two wrong-number calls tomorrow?

   (b) at least one wrong-number call tomorrow?

4. The number of alpha particles emitted by 1 gram of a radioactive element follows a Poisson distribution. On average, 1 gram of the element emits 3.9 alpha particles per second.

   What is the probability that, during the next second, the number of alpha particles emitted from 1 gram is

   (a) at most 6;

   (b) at least 2;

   (c) at least 3 and at most 6?

5. Suppose that there is, on average, one typographical error for every three pages of a book. If the number of typographical errors on a single page is a Poisson random variable, what is the probability of at least one error on a specific page of the book?

6. Suppose that earthquakes occur in a certain region according to a Poisson process at an average rate of 7 earthquakes per year.

   (a) What is the probability that no earthquakes occur in one year?

   (b) What is the probability that, in exactly three of the next eight years, no earthquakes occur?

7. In a small town, 12 accidents occurred during June, which has 30 days. Four of these accidents occurred on Friday, June 13.

   Assume that each accident is equally likely to occur on any day of the month, independently of the other accidents. What is the probability that at least 4 of the 12 accidents would occur on June 13?